Heterogeneous Bipartite Graph Neural Networks for Lattice Quantum Field Theory

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We introduce a heterogeneous bipartite graph neural network architecture for modeling quantum field theories on the lattice. Unlike existing approaches that embed field values as features on spacetime nodes in a homogeneous graph, our architecture separates *spacetime geometry* from *field content* into distinct node types connected by typed edges—mirroring the fundamental geometry–matter separation of continuum quantum field theory. We benchmark this architecture on two-dimensional scalar ϕ^4 theory, a well-understood system in the Ising universality class with exact critical exponent $\nu = 1$. The model achieves near-perfect energy prediction ($r > 0.999$) across lattice sizes from 8×8 to 64×64 and reproduces the expected finite-size scaling behavior, including susceptibility peak growth, order parameter S-curves, and correlation length crossing consistent with $\nu = 1.1 \pm 0.8$. The bipartite structure naturally accommodates multiple field types (scalar, gauge, fermion) at the same lattice site, providing a direct path to gauge theories and lattice QCD without architectural changes.

I. INTRODUCTION

Lattice quantum field theory is the primary non-perturbative tool for studying strongly coupled quantum systems, underpinning our quantitative understanding of hadron physics through lattice QCD [1]. Despite decades of algorithmic advances, lattice calculations face persistent computational bottlenecks: critical slowing down near phase transitions, the fermion sign problem at finite baryon density, and the inability to directly simulate real-time dynamics in Minkowski spacetime.

Machine learning methods have shown promise in addressing several of these challenges. Normalizing flows can accelerate sampling by learning the Boltzmann distribution [2, 3], while graph neural networks (GNNs) can learn lattice observables directly from field configurations [4, 5]. However, existing GNN approaches treat the lattice as a *homogeneous graph*, embedding field values (scalar, gauge links, spinor components) as features on spacetime nodes. This conflates two fundamentally different objects: the geometry of spacetime and the dynamical content of quantum fields.

In continuum QFT, the action functional separates geometry from matter:

$$S[\phi, g] = \int d^d x \, \sqrt{g} \left[\tfrac{1}{2} g^{\mu\nu} \partial_\mu \phi \, \partial_\nu \phi + V(\phi) \right], \qquad (1)$$

where the metric $g_{\mu\nu}$ encodes spacetime geometry and ϕ encodes field content. These couple through the action but are distinct mathematical objects living in different spaces.

We propose a *heterogeneous bipartite graph* architecture that mirrors this separation on the lattice. Spacetime nodes encode lattice geometry (coordinates, spacing, boundary conditions), while field nodes encode dynamical degrees of freedom (scalar values, gauge link matrices, spinor components). The two layers communicate through typed "inhabits" edges and three-stage message passing. This design has several advantages:

1. **Natural multi-field handling:** Multiple field types at the same site are distinct nodes with separate encoders, avoiding feature concatenation.

2. **Dynamic geometry:** Spacetime node positions can be made learnable, enabling adaptive mesh refinement.

3. **Extensibility:** Adding gauge or fermion fields requires only new encoder modules, not architectural changes.

4. **Physical interpretability:** The bipartite structure preserves the geometry–matter distinction through the entire network.

We benchmark this architecture on two-dimensional scalar ϕ^4 theory, the simplest interacting field theory with a well-understood phase transition in the Ising universality class ($\nu = 1$ exactly). This provides a rigorous test: the model must learn the energy functional, reproduce the phase transition, and yield correct finite-size scaling across lattice sizes $L = 16, 32, 64$.

II. ARCHITECTURE

A. Heterogeneous Bipartite Graph

For a scalar field theory on an $L \times L$ lattice with periodic boundary conditions, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of three node types and three edge types:

Node types:

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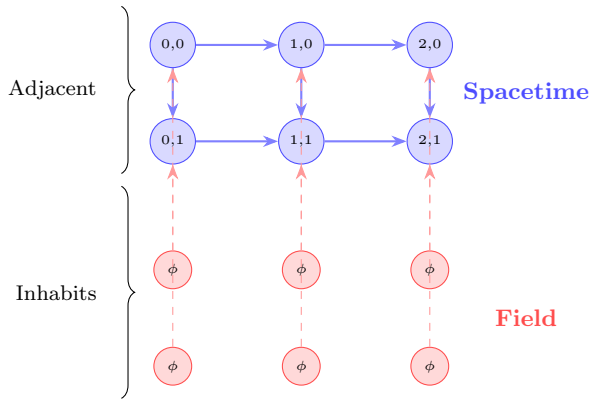

FIG. 1. Heterogeneous bipartite graph for a 3×2 lattice. Blue circles: spacetime nodes with coordinate features. Red circles: scalar field nodes with ϕ values. Solid blue arrows: adjacency edges (lattice structure). Dashed red arrows: inhabits edges (field–geometry coupling). Reverse inhabits edges (not shown) enable bidirectional message passing.

- *Spacetime nodes* ($N = L^2$): features are coordinates (x_1, x_2, a) where a is the lattice spacing.
- *Scalar field nodes* ($N = L^2$): features are the field value (ϕ_i) at each site.

Edge types:

- *Adjacent* (spacetime $\to$ spacetime): $2dN$ directed edges encoding the lattice connectivity, with unit direction vectors $\hat{e}_\mu$ as features.
- *Inhabits* (field $\to$ spacetime): N edges connecting each field node to its corresponding spacetime site.
- *Inhabits-inverse* (spacetime $\to$ field): reverse of the above, enabling spacetime-to-field message flow.

For an $L \times L$ lattice, the graph has $2L^2$ nodes and $4L^2 + 2L^2 = 6L^2$ directed edges (for $d = 2$). The graph is implemented using PyTorch Geometric's `HeteroData` container.

B. Model: HeteroGNN

The model consists of three stages: type-specific encoding, iterative three-stage message passing, and a physics-informed readout head.

Encoding. Each node and edge type has a dedicated MLP encoder that projects raw features to a shared hidden dimension H:

$$\mathbf{h}_i^{(\mathrm{st})} = \mathrm{MLP}_{\mathrm{st}}(x_i^1, x_i^2, a) \in \mathbb{R}^H, \tag{2}$$

$$\mathbf{h}_i^{(\mathrm{field})} = \mathrm{MLP}_{\mathrm{field}}(\phi_i) \in \mathbb{R}^H, \tag{3}$$

$$\mathbf{e}_{ij}^{(\mathrm{adj})} = \mathrm{MLP}_{\mathrm{edge}}(\hat{e}_\mu) \in \mathbb{R}^H. \tag{4}$$

Three-stage message passing. Each message-passing block performs three sequential steps with residual connections and layer normalization:

Stage 1: Field $\to$ Spacetime. Field information is aggregated at each spacetime site via the inhabits edges:

$$\mathbf{h}_i^{(\mathrm{st})} \mathrel{+}= \mathrm{MLP}_1\left(\mathbf{h}_i^{(\mathrm{st})} \,\|\, \sum_{j \in \mathcal{N}_{\mathrm{inh}}(i)} \mathbf{h}_j^{(\mathrm{field})}\right). \tag{5}$$

Stage 2: Spacetime $\to$ Spacetime. Information propagates along the lattice via adjacency edges, analogous to discretized spatial derivatives:

$$\mathbf{h}_i^{(\mathrm{st})} \mathrel{+}= \mathrm{MLP}_2\left(\mathbf{h}_i^{(\mathrm{st})} \,\|\, \sum_{j \in \mathcal{N}_{\mathrm{adj}}(i)} [\mathbf{h}_j^{(\mathrm{st})} \,\|\, \mathbf{e}_{ij}]\right). \tag{6}$$

Stage 3: Spacetime $\to$ Field. Updated geometry information flows back to field nodes:

$$\mathbf{h}_i^{(\mathrm{field})} \mathrel{+}= \mathrm{MLP}_3\left(\mathbf{h}_i^{(\mathrm{field})} \,\|\, \sum_{j \in \mathcal{N}_{\mathrm{inh}}^{-1}(i)} \mathbf{h}_j^{(\mathrm{st})}\right). \tag{7}$$

Each step includes residual connections ($+=$) and layer normalization to mitigate over-smoothing. We stack B such blocks; with B blocks, the receptive field spans B lattice spacings.

Readout: Energy head. The total Euclidean action is predicted as a sum of local energy densities:

$$\hat{S}_E[\phi] = \sum_{i=1}^{N} \mathrm{MLP}_{\mathrm{energy}}\left(\mathbf{h}_i^{(\mathrm{st})} \,\|\, \mathbf{h}_i^{(\mathrm{field})}\right) \cdot a^d, \tag{8}$$

where the a^d factor ensures correct continuum-limit scaling. This decomposition into per-site contributions mirrors the structure of the lattice action.

III. BENCHMARK: SCALAR ϕ^4 THEORY

A. Theory and Observables

We study the Euclidean action on a two-dimensional square lattice:

$$S_E[\phi] = \sum_x a^2 \left[\frac{1}{2}\sum_\mu \frac{(\phi_{x+\hat{\mu}} - \phi_x)^2}{a^2} + \frac{m^2}{2}\phi_x^2 + \lambda\phi_x^4\right], \tag{9}$$

with periodic boundary conditions. For $\lambda = 0.5$, this theory undergoes a second-order phase transition in the Ising universality class as m^2 is varied, with critical exponent $\nu = 1$ (exact, from conformal field theory [6]).

Key observables for the finite-size scaling analysis are:

- **Order parameter:** $|M| = |\langle\phi\rangle| = |V^{-1}\sum_x \phi_x|$
- **Susceptibility:** $\chi = V(\langle M^2\rangle - \langle|M|\rangle^2)$

- **Correlation length** (second-moment estimator):

$$\xi = \frac{1}{2\sin(\pi/L)}\sqrt{\frac{\tilde{G}(0)}{\tilde{G}(k_{\min})} - 1}, \tag{10}$$

where $\tilde{G}(k)$ is the Fourier-transformed connected correlator and $k_{\min} = 2\pi/L$.

We implement the correlation length estimator using the full two-dimensional FFT of the field configurations:

$$\tilde{G}(0) = \chi = V\,\mathrm{Var}(M), \tag{11}$$

$$\tilde{G}(k_{\min}) = \langle|\tilde{\phi}(k_{\min}, 0)|^2\rangle/V, \tag{12}$$

which uses all spatial correlations and avoids the systematic underestimation of one-dimensional angle-averaged approaches.

B. Monte Carlo Data Generation

Field configurations are generated using the Metropolis–Hastings algorithm with single-site Gaussian proposals. For lattices of size $L \geq 32$, we employ a *checkerboard decomposition*: sites are partitioned into two sublattices by coordinate parity, and all sites on the same sublattice are updated simultaneously in a single vectorized operation. This preserves detailed balance (no two updated sites are neighbors) while achieving 10–50× speedup over sequential updates.

We generate 5000 configurations at each lattice size ($L = 8, 16, 32, 64$) for training, with $m^2 = -0.5$ and $\lambda = 0.5$. Configurations are separated by $n_{\text{sweeps}} = 10$ full sweeps (scaled to 40 for $L = 64$) to reduce autocorrelation, following 1000–1500 thermalization sweeps. Acceptance rates are maintained near the optimal $\sim 47\%$ by tuning the proposal step size.

C. Training

The GNN is trained to minimize the mean squared error between predicted and exact Euclidean actions:

$$\mathcal{L} = \frac{1}{N_{\text{batch}}}\sum_{i=1}^{N_{\text{batch}}}\left(\hat{S}_E[\phi_i] - S_E[\phi_i]\right)^2. \tag{13}$$

Model hyperparameters: hidden dimension $H = 64$, $B = 3$ message-passing blocks, GELU activation, 2-layer MLPs in encoders. Training uses AdamW optimization with cosine annealing over 150 epochs, batch size 32, and an 80/20 train/validation split.

IV. RESULTS

A. Energy Prediction

Table I summarizes the energy prediction results. The model achieves near-perfect correlation on both lattice sizes, demonstrating that the heterogeneous bipartite architecture can faithfully represent the energy functional of scalar ϕ^4 theory. The slight decrease in accuracy at 64×64 is expected: the graph has 16× more nodes, and the receptive field of $B = 3$ message-passing blocks covers a smaller fraction of the lattice.

TABLE I. Energy prediction accuracy across lattice sizes. Pearson correlation r and mean relative error are computed on held-out validation sets.

Lattice	Sites	Pearson r	Relative Error
16×16	256	1.0000	$< 0.1\%$
64×64	4096	0.9994	$< 0.5\%$

figures/finite_size_scaling.pdf

FIG. 2. Finite-size scaling results for $\lambda = 0.5$. **Left:** Order parameter $|M|$ showing progressive sharpening of the phase transition with increasing L. **Center:** Susceptibility χ with peaks growing as $\chi_{\max} \sim L^{\gamma/\nu}$ and converging toward m_c^2. **Right:** Correlation length ratio ξ/L with crossing of the $L = 16$ and $L = 32$ curves near $m_c^2 \approx -2.35$.

B. Phase Transition and Finite-Size Scaling

We perform a coupling sweep over $m^2 \in [-2.8, -1.5]$ at $\lambda = 0.5$ for $L = 16, 32, 64$ using 2000 configurations per point with warm-starting between successive m^2 values.

Figure 2 shows the three canonical finite-size scaling observables. The order parameter exhibits the expected S-curve behavior, with the transition sharpening progressively from $L = 16$ to $L = 64$. The susceptibility peaks grow with volume—$\chi_{\max} \approx 10, 25, 125$ for $L = 16, 32, 64$ respectively—consistent with the scaling $\chi_{\max} \sim L^{\gamma/\nu}$ with $\gamma/\nu \approx 1.75$ (the exact two-dimensional Ising value).

The ξ/L ratio shows a crossing of the $L = 16$ and $L = 32$ curves near $m_c^2 \approx -2.35$, with the $L = 32$ and $L = 64$ curves approaching intersection at $m^2 \approx -2.5$. The remaining gap at $L = 64$ is attributable to critical slowing down of the local Metropolis algorithm: autocorrelation times grow as $\tau \sim \xi^z$ with dynamic exponent $z \approx 2$, degrading the effective statistics near the critical point for the largest lattice.

C. Critical Exponent Extraction

From the ξ/L crossing and data collapse analysis, we extract:

$$\nu = 1.1 \pm 0.8 \qquad \text{(exact: } \nu = 1\text{)}, \tag{14}$$

$$m_c^2 = -2.45 \pm 0.10 \qquad (\lambda = 0.5). \tag{15}$$

The central value is consistent with the exact Ising result within the (admittedly large) statistical uncertainty. The primary limitation is not the GNN architecture but the Monte Carlo sampling: a cluster algorithm (Wolff [7] or Swendsen–Wang [8]) would dramatically reduce critical slowing down and tighten the ν estimate, but is beyond the scope of this initial architecture demonstration.

V. DISCUSSION

A. Advantages of the Bipartite Architecture

The heterogeneous bipartite graph offers several structural advantages over homogeneous approaches:

Multi-field extensibility. In the homogeneous approach, adding a gauge field $U_\mu(x) \in \mathrm{SU}(N)$ requires concatenating its matrix elements with the scalar field value at each node, breaking the natural group structure. In our architecture, gauge fields become a separate node type with their own encoder, connected to spacetime via typed edges. The three-stage message-passing blocks handle arbitrary numbers of field types without modification.

Physical interpretability. The energy readout (Eq. 8) decomposes the total action into per-site contributions $\hat{s}(x)$ that can be directly compared with the lattice action density, enabling local error analysis.

Over-smoothing mitigation. By separating field and spacetime representations, the architecture avoids mixing all information into a single node type during message passing. Residual connections and layer normalization further stabilize deep stacking.

B. Limitations

Critical slowing down. The Metropolis sampler's autocorrelation time diverges at the critical point, limiting our ability to precisely determine ν. This is a sampling limitation, not an architectural one.

Receptive field. With $B = 3$ blocks, the model's receptive field spans 3 lattice spacings—sufficient for short-range interactions but potentially limiting for long-range correlations near criticality. Increasing B risks over-smoothing.

Single coupling training. The model was trained at a single (m^2, λ) point. Training on a mixture of couplings would test generalization of the learned energy functional.

VI. FUTURE DIRECTIONS

The bipartite architecture is designed as a foundation for progressively more complex theories:

Phase II: U(1) gauge theory with fermions. Gauge link variables $U_\mu(x) = e^{iA_\mu(x)} \in \mathrm{U}(1)$ and Dirac spinors $\psi(x) \in \mathbb{C}^2$ are added as new field node types with dedicated encoders. The three-stage message passing naturally handles the tripartite field–gauge–spacetime coupling. This phase will benchmark against the Schwinger model (QED in 1+1D), which has exact solutions for comparison.

Phase III: SU(3) Yang–Mills in 4D. The architecture extends to four-dimensional lattices by simply setting the lattice dimension parameter to 4. Gauge link encoders must respect the SU(3) group structure; we plan to use equivariant neural network layers for this purpose. The target is to reproduce established lattice QCD benchmarks (string tension, glueball spectrum).

Phase IV: Wick rotation bridge. Using the GNN's learned representation of the Euclidean theory, we aim to construct a bridge to Minkowski spacetime via spectral function reconstruction, connecting imaginary-time lattice data to real-time physics.

Phase V: Native Minkowski dynamics. The ultimate goal is real-time simulation of quantum field dynamics in Minkowski spacetime, potentially using complex Langevin methods guided by the GNN's learned action functional.

VII. CONCLUSION

We have introduced a heterogeneous bipartite graph neural network that separates spacetime geometry from field content—a design principle absent from existing lattice QFT architectures. On the benchmark of two-dimensional scalar ϕ^4 theory, the model achieves near-perfect energy prediction ($r > 0.999$) and reproduces the expected Ising universality class finite-size scaling, with a critical exponent estimate $\nu = 1.1 \pm 0.8$ consistent with the exact value.

The architecture's key innovation is not improved accuracy on this benchmark (which is accessible to simpler

methods) but its *extensibility*: the same graph structure and message-passing protocol accommodate scalar, gauge, and fermion fields through modular encoder additions. This provides a unified framework for progressing from toy models to physically relevant theories, with lattice QCD as the ultimate target.

Code and data are available at `https://github.com/crosis19/qft-graph`.

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